

Assessment Task 1 (AT1) – Part 2

DevSecOps Risk Detective Challenge

Scenario-Based Collaborative Analysis

Overview

This collaborative assessment evaluates students' ability to apply foundational DevSecOps concepts to realistic software engineering scenarios. Each team acts as an independent DevSecOps consulting firm engaged by an organization to identify security risks, analyze vulnerabilities, recommend OWASP-aligned improvements, and explain the business value of integrating security into the software development lifecycle.

Instructions

- Work in teams of 4–5 students.
- Each team receives one unique industry case study.
- Analyze the scenario and complete all investigation tasks.
- Prepare a 5-minute presentation and one consolidated report.
- Support every recommendation with technical justification.

Investigation Tasks (Applicable to All Teams)

Existing Process Review (5 Marks)

Identify at least five weaknesses in the organization's current software development process.

Threat Identification (5 Marks)

Identify possible cyber threats and explain how they may exploit the environment.

Vulnerability Assessment (5 Marks)

Identify vulnerabilities, classify them (High/Medium/Low) and justify each classification.

DevOps vs DevSecOps Analysis (5 Marks)

Prepare a comparison table with at least eight points.

OWASP Recommendations (4 Marks)

Recommend five OWASP best practices and explain the benefit of each.

Business Impact Analysis (3 Marks)

Explain financial, operational, legal, compliance and reputational impact.

Presentation (3 Marks)

Present findings and answer faculty questions.

Team 1 – Digital Banking Platform

Background

SecureBank Ltd. operates a nationwide digital banking platform serving more than five million customers. The bank offers online banking, mobile banking, digital payments, loan processing and investment services. To remain competitive with fintech companies, management mandates weekly software releases.

The development team commits directly to the main branch, code reviews are optional, security testing is performed only before production deployment, third-party libraries are adopted without verification, password policies are weak, and API authentication is inconsistent. Application logs are generated but rarely reviewed. Developers have not received secure coding training.

Recently, customers reported unauthorized login attempts and suspicious transactions. Internal investigation revealed exposed API keys, delayed security patches and outdated libraries.

Investigation Questions

1. Identify at least five security risks.
2. Identify at least five vulnerabilities and classify each as High, Medium or Low.
3. List possible cyber threats and explain their impact.
4. Prepare a DevOps vs DevSecOps comparison table (minimum eight points).
5. Recommend five OWASP security best practices.
6. Explain the business impact if these issues remain unresolved.
7. Create a concept map linking Risk → Vulnerability → DevSecOps Practice → Business Benefit.
8. Prepare a five-minute presentation summarizing your findings.

Team 2 – E-Commerce Platform

Background

ShopNow is a fast-growing online marketplace processing thousands of customer orders every day. Marketing campaigns require rapid feature delivery, so developers frequently deploy directly to production. Peer reviews are skipped, secure coding standards are not enforced, payment services are integrated with multiple third-party vendors, and vulnerability assessments are performed only after incidents.

Customers have recently complained about fraudulent purchases and unexpected account activity. Management suspects weaknesses in authentication, session handling and application security.

Investigation Questions

1. Identify at least five security risks.
2. Identify at least five vulnerabilities and classify each as High, Medium or Low.
3. List possible cyber threats and explain their impact.
4. Prepare a DevOps vs DevSecOps comparison table (minimum eight points).
5. Recommend five OWASP security best practices.
6. Explain the business impact if these issues remain unresolved.
7. Create a concept map linking Risk → Vulnerability → DevSecOps Practice → Business Benefit.
8. Prepare a five-minute presentation summarizing your findings.

Team 3 – Hospital Information System

Background

CarePlus Hospital provides an online patient portal containing electronic medical records, prescriptions, laboratory reports and appointment scheduling. Weekly updates are released to improve patient experience. Passwords are stored using weak practices, security reviews are skipped to meet deadlines, and staff access the system from unmanaged devices.

A recent audit found unauthorized access attempts against patient records. Hospital management is concerned about patient privacy, regulatory compliance and service availability.

Investigation Questions

1. Identify at least five security risks.
2. Identify at least five vulnerabilities and classify each as High, Medium or Low.
3. List possible cyber threats and explain their impact.
4. Prepare a DevOps vs DevSecOps comparison table (minimum eight points).
5. Recommend five OWASP security best practices.
6. Explain the business impact if these issues remain unresolved.
7. Create a concept map linking Risk → Vulnerability → DevSecOps Practice → Business Benefit.
8. Prepare a five-minute presentation summarizing your findings.

Team 4 – University ERP Portal

Background

A university ERP system manages admissions, examinations, attendance, fee payments and student records for over 20,000 students. Development focuses primarily on new functionality. Password reset links never expire, uploaded files are not validated, security testing occurs only after deployment and user activity logs are rarely reviewed.

The administration recently discovered unusual changes in student records and suspects weaknesses in access control and authentication.

Investigation Questions

1. Identify at least five security risks.
2. Identify at least five vulnerabilities and classify each as High, Medium or Low.
3. List possible cyber threats and explain their impact.
4. Prepare a DevOps vs DevSecOps comparison table (minimum eight points).
5. Recommend five OWASP security best practices.
6. Explain the business impact if these issues remain unresolved.
7. Create a concept map linking Risk → Vulnerability → DevSecOps Practice → Business Benefit.
8. Prepare a five-minute presentation summarizing your findings.

Team 5 – Airline Reservation System

Background

SkyFly Airlines continuously updates its online booking platform with promotional offers and new services. The application depends heavily on open-source libraries but no formal process exists for checking known vulnerabilities. Patches are delayed to avoid interrupting business operations, API protection is inconsistent and security reviews are reactive.

A recent outage disrupted online ticket bookings, causing revenue loss and customer complaints.

Investigation Questions

1. Identify at least five security risks.
2. Identify at least five vulnerabilities and classify each as High, Medium or Low.
3. List possible cyber threats and explain their impact.
4. Prepare a DevOps vs DevSecOps comparison table (minimum eight points).
5. Recommend five OWASP security best practices.
6. Explain the business impact if these issues remain unresolved.
7. Create a concept map linking Risk → Vulnerability → DevSecOps Practice → Business Benefit.
8. Prepare a five-minute presentation summarizing your findings.

Team 6 – Government Citizen Services Portal

Background

A national citizen services portal provides tax filing, identity services, certificate applications and public grievance management. Different development teams build independent modules using inconsistent security practices. Identity management is weak, security policies are documented but rarely enforced, logging is limited and monitoring is largely manual.

An external audit highlighted several compliance concerns and recommended immediate security improvements.

Investigation Questions

1. Identify at least five security risks.
2. Identify at least five vulnerabilities and classify each as High, Medium or Low.
3. List possible cyber threats and explain their impact.
4. Prepare a DevOps vs DevSecOps comparison table (minimum eight points).
5. Recommend five OWASP security best practices.
6. Explain the business impact if these issues remain unresolved.
7. Create a concept map linking Risk → Vulnerability → DevSecOps Practice → Business Benefit.
8. Prepare a five-minute presentation summarizing your findings.

Team 7 – Social Media Platform

Background

ConnectHub is a rapidly expanding social networking platform supporting millions of users who upload photos, videos and messages daily. The platform exposes multiple public APIs to mobile applications and third-party partners. Access controls differ across services, suspicious user behaviour is rarely monitored and the security team is involved only after incidents.

Recent reports indicate fake accounts, unauthorized data access and leaked user information, raising concerns about privacy and trust.

Investigation Questions

1. Identify at least five security risks.
2. Identify at least five vulnerabilities and classify each as High, Medium or Low.
3. List possible cyber threats and explain their impact.
4. Prepare a DevOps vs DevSecOps comparison table (minimum eight points).
5. Recommend five OWASP security best practices.
6. Explain the business impact if these issues remain unresolved.
7. Create a concept map linking Risk → Vulnerability → DevSecOps Practice → Business Benefit.
8. Prepare a five-minute presentation summarizing your findings.

Evaluation Rubric (30 Marks)

Criteria	Marks
Identification of Security Risks	5
Threat & Vulnerability Analysis	5
DevOps vs DevSecOps Analysis	5
OWASP Recommendations	4
Business Impact Analysis	3
Technical Justification	3
Team Collaboration	2
Presentation	3